Solutions for CES 552 Assignment # 3

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1. p.576 #29 The next window size is 1KB, then 2KB for the second transmission, 4KB for the third transmission, and 8KB for the fourth transmission. After the fourth transmission, the window size is 9KB due to the threshold.

2. p.576 #30

$$RTT = 0.9 \times 30 + (1 - 0.9) \times 26 = 29.6 \quad (1)$$
$$RTT = 0.9 \times 29.6 + (1 - 0.9) \times 32 = 29.84 \quad (2)$$
$$RTT = 0.9 \times 29.84 + (1 - 0.9) \times 24 = 29.256 \quad (3)$$

Solutions for CES 552 Assignment # 4

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1. p.576 #31
   It takes

$$t = 65,535 \times 8/10^9 \approx 0.524\text{ms} \tag{1}$$

   for the host to put the entire packet in the channel.

   The round trip delay is $2 \times 10 = 20$ ms.

   Therefor, the channel is used for 0.524 ms every 20 ms. The line efficiency is $0.524/20 = 2.6\%$

2. p.577 #40
   The round trip propagation delay is

$$RTT = 2 \times \frac{2 \times 10^4}{3 \times 10^8} = 133.3\mu s \tag{2}$$

   If the RTT equal to the transmission delay of a 1KB packet, then

$$R = 1024 \times 8/RTT = 61.46Mbps \tag{3}$$

3. p. 716 #5
   Even though there is no retransmission on transport layer, the DNS program at the application layer will do the retransmission. Every time a DNS request packet is sent out, the process on application layer will start a timer. If the timer expires before it gets a reply, it will retransmit the request packet.